DINITHI DISSANAYAKE

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PhD researcher developing multimodal AI for wearable and egocentric systems to understand human attention, cognitive and affective states through vision, gaze, audio, and physiological sensing. My research spans representation learning, multimodal fusion, efficient on-device inference, and human-centered evaluation.

EDUCATION

PhD in Computing *Aug 2023 – Present*
School of Computing, National University of Singapore, Singapore
Advisor: Prof. Suranga Nanayakkara

BSc in Engineering, Electronics and Telecommunication Engineering *Aug 2017 – Jul 2022*
University of Moratuwa, Sri Lanka
First Class Honours | GPA: 3.92/4.20 | Dean's List: 7/8 semesters

SELECTED RESEARCH PROJECTS

HeadRoom: Inferring Sensory Channel Availability for Wearable AI *UIST 2026*

- Developed lightweight estimators of moment-to-moment visual and auditory availability to route notifications to the less-occupied sensory channel.
- Deployed inference on Meta Quest 3S using Unity and ONNX (10.98 ms latency; 10 MB memory overhead).
- Evaluated response time in a controlled user study ($N = 22$; 796 probes); HeadRoom was significantly faster.

Egocentric Gaze Anticipation *Ongoing*

- Developing spatial–temporal models to predict *when* a major gaze shift begins and *where* gaze will land, from gaze direction, head pose, and optional egocentric video.
- Investigating neuroscience-inspired spatial and temporal processing using lightweight gaze/head-motion signals and learned video representations.
- Building event-focused evaluation protocols for large gaze shifts across Ego4D, Aria, and HDEpic datasets.

Understanding Human Oversight of AI Coding Agents *Under review*

- Designed a controlled AI-assisted programming study ($N = 36$) contrasting correct implementations, visibly failing defects, and silent defects that passed initial checks.
- Analyzed interaction logs, gaze, and physiology to distinguish task-relevant evaluation, defect discovery, and resolution from superficial acceptance.
- Found high confidence despite oversight failures: 79% of missed defects received confidence ratings of 4–5/5.

Multimodal Emotion Mimicry Intensity Prediction *CVPR Workshops 2026*

- Developed a two-stage framework to regress six emotion-mimicry intensities from in-the-wild clips using textual, acoustic, and visual representations, with an optional motion branch.
- Trained modality-specific encoders before fusing representations through a lightweight regressor with modality dropout and controlled encoder adaptation.
- Placed 3rd in the Hume-ABAW10 EMI Challenge (average test Pearson correlation $r = 0.57$).

VRSense: Modeling and Explaining VR Cybersickness *CHI 2025*

- Developed multimodal prediction and explainability workflows linking headset motion, eye tracking, and physiological measurements to subjective cybersickness during VR gameplay.
- Translated model explanations into actionable design considerations through interviews with VR developers.

HONORS & AWARDS

Distinguished Poster Award – Meta Aria Research Summit for Egocentric Intelligence (HeadRoom) *2026*
3rd Place, Emotion Mimicry Intensity Prediction Challenge – CVPR Workshops *2026*
Doctoral Symposium Participant – ACM UIST *2026*

TECHNICAL SKILLS

ML & Programming: Python, C++, SQL, PyTorch, TensorFlow, OpenCV
Modeling: Multimodal fusion, egocentric vision, temporal forecasting, self-supervised learning, affect recognition, 3D vision
Sensing & XR: Meta Aria, Meta Quest, Tobii, Shimmer; gaze, IMU, PPG, EDA, HRV
Deployment & Tools: ONNX, Unity, Docker, Git, AWS, GCP; on-device inference

SELECTED PUBLICATIONS

HeadRoom: Lightweight, Edge-deployable Pipeline for Adaptive Notification Routing. *UIST '26*.
Dinithi Dissanayake, Prasanth Sasikumar, Suranga Nanayakkara

Two-Stage Multimodal Framework for Emotion Mimicry Intensity Prediction. *CVPR Workshops '26*.
Dinithi Dissanayake, Shaveen Silva, Ovindu Atukorala, Prasanth Sasikumar, Suranga Nanayakkara

Interrupting Autopilot: Evaluating Sensory Interventions for Momentary Self-Regulation. *CHI EA '26*.
Dinithi Dissanayake, Muftee Mysan, Yifei Luo, Timothy Antoni, Peter Cleveland, Prasanth S., Suranga Nanayakkara

VRSense: An Explainable System to Help Mitigate Cybersickness in VR Games. *CHI EA '25*.
Dinithi Dissanayake, Chitralekha Gupta, Prasanth Sasikumar, Suranga Nanayakkara

Navigating the State of Cognitive Flow: Context-Aware AI Interventions. *CHI '25 Workshops*.
Dinithi Dissanayake, Suranga Nanayakkara

CrossPoint: Self-Supervised Cross-Modal Contrastive Learning for 3D Point Cloud Understanding. *CVPR '22*.
Aflal Afham, Isuru Dissanayake, **Dinithi Dissanayake**, Amaya Dharmasiri, Kanchana Thilakarathna, Ranga Rodrigo

3DLatNav: Navigating Generative Latent Spaces for Semantic-Aware 3D Object Manipulation. *ECCV Workshops '22*.
Amaya Dharmasiri, **Dinithi Dissanayake**, Isuru Dissanayake, Aflal Afham, Ranga Rodrigo, Kanchana Thilakarathna

MANUSCRIPTS UNDER REVIEW

Looks Fine to Me: Understanding Oversight in Agentic Interactions.
Dinithi Dissanayake, Yasith S., Hyung Lee, Yadeesha Weerasinghe, Sanjay Saha, Pattie Maes, Suranga Nanayakkara

A Systematic Literature Review of User-Aware Adaptive Assistive Wearables.
Dinithi Dissanayake, Hyong Woon Lee, Moritz Messerschmidt, Yize Wei, Suranga Nanayakkara

RESEARCH & INDUSTRY EXPERIENCE

Research Collaboration, Meta Reality Labs × NUS *Aug 2024 – Feb 2025*

- Led data collection and synchronization for a VR gameplay study ($N = 150$), coordinating multimodal sensor pipelines and labeled sessions to support cybersickness modeling research.

Data Analytics Consultant, LIRNEasia *Jan 2023 – Aug 2023 (Part-time)*

- Built satellite-imagery classification and geospatial analysis workflows using QGIS and Google Earth Engine to support an urbanization index for Sri Lanka.

Data Engineer, Axiata Digital Labs *May 2022 – Aug 2023*

- Engineered ingestion and ML deployment pipelines supporting computer vision, forecasting, and customer-churn analytics on an internal AI platform.

TEACHING & VOLUNTEERING SERVICE

Mentored research interns on physiological wearable-data analysis, NUS Augmented Human Lab.
Teaching Assistant, CS2100 Computer Organisation, National University of Singapore.
Reviewer: CHI'26, 27; DIS'26; UIST'26
Student Volunteer chair, Augmented Humans (AHs) 2027